

APPENDIX A

PROCESSOR INSTRUCTION SUMMARY

This appendix provides a quick reference of the M68000 family instructions. The organization of this section is by processors and their addressing modes. All references to the MC68000, MC68020, and MC68030 include references to the corresponding embedded controllers, MC68EC000, MC68EC020, and MC68EC030. All references to the MC68040 include the MC68LC040 and MC68EC040. This referencing applies throughout this section unless otherwise specified. Table A-1 lists the M68000 family instructions by mnemonic and indicates which processors they apply to.

Table A-1. M68000 Family Instruction Set And Processor Cross-Reference

Mnemonic	68000	68008	68010	68020	68030	68040	68881/ 68882	68851	CPU32
ABCD	X	X	X	X	X	X			X
ADD	X	X	X	X	X	X			X
ADDA	X	X	X	X	X	X			X
ADDI	X	X	X	X	X	X			X
ADDQ	X	X	X	X	X	X			X
ADDX	X	X	X	X	X	X			X
AND	X	X	X	X	X	X			X
ANDI	X	X	X	X	X	X			X
ANDI to CCR	X	X	X	X	X	X			X
ANDI to SR ¹	X	X	X	X	X	X			X
ASL, ASR	X	X	X	X	X	X			X
Bcc	X	X	X	X	X	X			X
BCHG	X	X	X	X	X	X			X
BCLR	X	X	X	X	X	X			X
BFCHG				X	X	X			
BFCLR				X	X	X			
BFEXTS				X	X	X			
BFEXTU				X	X	X			
BFFFO				X	X	X			

Table A-1. M68000 Family Instruction Set And Processor Cross-Reference (Continued)

Mnemonic	68000	68008	68010	68020	68030	68040	68881/ 68882	68851	CPU32
BFINS				X	X	X			
BFSET				X	X	X			
BFTST				X	X	X			
BGND									X
BKPT			X	X	X	X			X
BRA	X	X	X	X	X	X			X
BSET	X	X	X	X	X	X			X
BSR	X	X	X	X	X	X			X
BTST	X	X	X	X	X	X			X
CALLM				X					
CAS, CAS2				X	X	X			
CHK	X	X	X	X	X	X			X
CHK2				X	X	X			X
CINV ¹						X			
CLR	X	X	X	X	X	X			X
CMP	X	X	X	X	X	X			X
CMPA	X	X	X	X	X	X			X
CMPI	X	X	X	X	X	X			X
CMPM	X	X	X	X	X	X			X
CMP2				X	X	X			X
cpBcc				X	X				
cpDBcc				X	X				
cpGEN				X	X				
cpRESTORE ¹				X	X				
cpSAVE ¹				X	X				
cpScc				X	X				
cpTRAPcc				X	X				
CPUSH ¹						X			
DBcc	X	X	X	X	X	X			X
DIVS	X	X	X	X	X	X			X
DIVSL				X	X	X			X
DIVU	X	X	X	X	X	X			X
DIVUL				X	X	X			X

**Table A-1. M68000 Family Instruction Set And
Processor Cross-Reference (Continued)**

Mnemonic	68000	68008	68010	68020	68030	68040	68881/ 68882	68851	CPU32
EOR	X	X	X	X	X	X			X
EORI	X	X	X	X	X	X			X
EORI to CCR	X	X	X	X	X	X			X
EORI to SR ¹	X	X	X	X	X	X			X
EXG	X	X	X	X	X	X			X
EXT	X	X	X	X	X	X			X
EXTB				X	X	X			X
FABS						X ²	X		
FSABS, FDABS						X ²			
FACOS						2,3	X		
FADD						X ²	X		
FSADD, FDADD						X ²			
FASIN						2,3	X		
FATAN						2,3	X		
FATANH						2,3	X		
FBcc						X ²	X		
FCMP						X ²	X		
FCOS						2,3	X		
FCOSH						2,3	X		
FDBcc						X ²	X		
FDIV						X ²	X		
FSDIV, FDDIV						X ²			
FETOX						2,3	X		
FETOXM1						2,3	X		
FGETEXP						2,3	X		
FGETMAN						2,3	X		
FINT						2,3	X		
FINTRZ						2,3	X		
FLOG10						2,3	X		
FLOG2						2,3	X		
FLOGN						2,3	X		

Table A-1. M68000 Family Instruction Set And Processor Cross-Reference (Continued)

Mnemonic	68000	68008	68010	68020	68030	68040	68881/ 68882	68851	CPU32
FLOGNP1						2,3			
FMOD						2,3	X		
FMOVE						X ²	X		
FSMOVE, FDMOVE						X ²			
FMOVECR						2,3	X		
FMOVEM						X ²	X		
FMUL						X ²	X		
FSMUL, FDMUL						X ²			
FNEG						X ²	X		
FSNEG, FDNEG						X ²			
FNOP						X ²	X		
FREM						2,3	X		
FRESTORE ¹						X ²	X		
FSAVE*						X ²	X		
FSCALE						2,3	X		
FScc						X ²	X		
FSGLDIV						2,3	X		
FSGLMUL						2,3	X		
FSIN						2,3	X		
FSINCOS						2,3	X		
FSINH						2,3	X		
FSQRT						X ²	X		
FSSQRT, FDSQRT						X ²			
FSUB						X ²	X		
FSSUB, FDSUB						X ²			
FTAN						2,3	X		
FTANH						2,3	X		
FTENTOX						2,3	X		
FTRAPcc						X ²	X		
FTST						X ²	X		

**Table A-1. M68000 Family Instruction Set And
Processor Cross-Reference (Continued)**

Mnemonic	68000	68008	68010	68020	68030	68040	68881/ 68882	68851	CPU32
FTWOTOX						2,3	X		
ILLEGAL	X	X	X	X	X	X			X
JMP	X	X	X	X	X	X			X
JSR	X	X	X	X	X	X			X
LEA	X	X	X	X	X	X			X
LINK	X	X	X	X	X	X			X
LPSTOP									X
LSL,LSR	X	X	X	X	X	X			X
MOVE	X	X	X	X	X	X			X
MOVEA	X	X	X	X	X	X			X
MOVE from CCR			X	X	X	X			X
MOVE to CCR	X	X	X	X	X	X			X
MOVE from SR ¹	4	4	X	X	X	X			X
MOVE to SR ¹	X	X	X	X	X	X			X
MOVE USP ¹	X	X	X	X	X	X			X
MOVE16						X			
MOVEC ¹			X	X	X	X			X
MOVEM	X	X	X	X	X	X			X
MOVEP	X	X	X	X	X	X			X
MOVEQ	X	X	X	X	X	X			X
MOVES ¹			X	X	X	X			X
MULS	X	X	X	X	X	X			X
MULU	X	X	X	X	X	X			X
NBCD	X	X	X	X	X	X			X
NEG	X	X	X	X	X	X			X
NEGX	X	X	X	X	X	X			X
NOP	X	X	X	X	X	X			X
NOT	X	X	X	X	X	X			X
OR	X	X	X	X	X	X			X

Table A-1. M68000 Family Instruction Set And Processor Cross-Reference (Continued)

Mnemonic	68000	68008	68010	68020	68030	68040	68881/ 68882	68851	CPU32
ORI	X	X	X	X	X	X			X
ORI to CCR	X	X	X	X	X	X			X
ORI to SR ¹	X	X	X	X	X	X			X
PACK				X	X	X			
PBcc ¹								X	
PDBcc ¹								X	
PEA	X	X	X	X	X	X			X
PFLUSH ¹					X ⁵	X		X	
PFLUSHA ¹					X ⁵			X	
PFLUSHR ¹								X	
PFLUSHS ¹								X	
PLOAD ¹					X ⁵			X	
PMOVE ¹					X			X	
PRESTORE ¹								X	
PSAVE ¹								X	
PScc ¹								X	
PTEST ¹					X	X		X	
PTRAPcc ¹								X	
PVALID								X	
RESET ¹	X	X	X	X	X	X			X
ROL,ROR	X	X	X	X	X	X			X
ROXL, ROXR	X	X	X	X	X	X			X
RTD			X	X	X	X			X
RTE ¹	X	X	X	X	X	X			X
RTM				X					
RTR	X	X	X	X	X	X			X
RTS	X	X	X	X	X	X			X
SBCD	X	X	X	X	X	X			X
Scc	X	X	X	X	X	X			X
STOP ¹	X	X	X	X	X	X			X
SUB	X	X	X	X	X	X			X
SUBA	X	X	X	X	X	X			X
SUBI	X	X	X	X	X	X			X
SUBQ	X	X	X	X	X	X			X
SUBX	X	X	X	X	X	X			X

**Table A-1. M68000 Family Instruction Set And
Processor Cross-Reference (Concluded)**

Mnemonic	68000	68008	68010	68020	68030	68040	68881/ 68882	68851	CPU32
SWAP	X	X	X	X	X	X			X
TAS	X	X	X	X	X	X			X
TBLS, TBLSN									X
TBLU, TBLUN									X
TRAP	X	X	X	X	X	X			X
TRAPcc				X	X	X			X
TRAPV	X	X	X	X	X	X			X
TST	X	X	X	X	X	X			X
UNLK	X	X	X	X	X	X			X
UNPK				X	X	X			

NOTES:

1. Privileged (Supervisor) Instruction.
2. Not applicable to MC68EC040 and MC68LC040
3. These instructions are software supported on the MC68040.
4. This instruction is not privileged for the MC68000 and MC68008.
5. Not applicable to MC68EC030.

Table A-2 lists the M68000 family instructions by mnemonics, followed by the descriptive name.

Table A-2. M68000 Family Instruction Set

Mnemonic	Description
ABCD	Add Decimal with Extend
ADD	Add
ADDA	Address
ADDI	Add Immediate
ADDQ	Add Quick
ADDX	Add with Extend
AND	Logical AND
ANDI	Logical AND Immediate
ANDI to CCR	AND Immediate to Condition Code Register
ANDI to SR	AND Immediate to Status Register
ASL, ASR	Arithmetic Shift Left and Right
Bcc	Branch Conditionally
BCHG	Test Bit and Change
BCLR	Test Bit and Clear
BFCHG	Test Bit Field and Change
BFCLR	Test Bit Field and Clear
BFEXTS	Signed Bit Field Extract
BFEXTU	Unsigned Bit Field Extract
BFFFO	Bit Field Find First One
BFINS	Bit Field Insert
BFSET	Test Bit Field and Set
BFTST	Test Bit Field
BGND	Enter Background Mode
BKPT	Breakpoint
BRA	Branch
BSET	Test Bit and Set
BSR	Branch to Subroutine
BTST	Test Bit
CALLM	CALL Module
CAS	Compare and Swap Operands
CAS2	Compare and Swap Dual Operands
CHK	Check Register Against Bound
CHK2	Check Register Against Upper and Lower Bounds
CINV	Invalidate Cache Entries
CLR	Clear
CMP	Compare
CMPA	Compare Address
CMPI	Compare Immediate
CMPM	Compare Memory to Memory
CMP2	Compare Register Against Upper and Lower Bounds
cpBcc	Branch on Coprocessor Condition
cpDBcc	Test Coprocessor Condition Decrement and Branch
cpGEN	Coprocessor General Function
cpRESTORE	Coprocessor Restore Function

Table A-2. M68000 Family Instruction Set (Continued)

Mnemonic	Description
cpSAVE	Coprocessor Save Function
cpScc	Set on Coprocessor Condition
cpTRAPcc	Trap on Coprocessor Condition
CPUSH	Push then Invalidate Cache Entries
DBcc	Test Condition, Decrement and Branch
DIVS, DIVSL	Signed Divide
DIVU, DIVUL	Unsigned Divide
EOR	Logical Exclusive-OR
EORI	Logical Exclusive-OR Immediate
EORI to CCR	Exclusive-OR Immediate to Condition Code Register
EORI to SR	Exclusive-OR Immediate to Status Register
EXG	Exchange Registers
EXT, EXTB	Sign Extend
FABS	Floating-Point Absolute Value
FSFABS, FDFABS	Floating-Point Absolute Value (Single/Double Precision)
FACOS	Floating-Point Arc Cosine
FADD	Floating-Point Add
FSADD, FDADD	Floating-Point Add (Single/Double Precision)
FASIN	Floating-Point Arc Sine
FATAN	Floating-Point Arc Tangent
FATANH	Floating-Point Hyperbolic Arc Tangent
FBcc	Floating-Point Branch
FCMP	Floating-Point Compare
FCOS	Floating-Point Cosine
FCOSH	Floating-Point Hyperbolic Cosine
FDBcc	Floating-Point Decrement and Branch
FDIV	Floating-Point Divide
FSDIV, FDDIV	Floating-Point Divide (Single/Double Precision)
FETOX	Floating-Point ex
FETOXM1	Floating-Point ex - 1
FGETEXP	Floating-Point Get Exponent
FGETMAN	Floating-Point Get Mantissa
FINT	Floating-Point Integer Part
FINTRZ	Floating-Point Integer Part, Round-to-Zero
FLOG10	Floating-Point Log10
FLOG2	Floating-Point Log2
FLOGN	Floating-Point Loge
FLOGNP1	Floating-Point Loge (x + 1)
FMOD	Floating-Point Modulo Remainder
FMOVE	Move Floating-Point Register
FSMOVE, FDMOVE	Move Floating-Point Register (Single/Double Precision)
FMOVECR	Move Constant ROM
FMOVEM	Move Multiple Floating-Point Registers
FMUL	Floating-Point Multiply
FSMUL, FDMUL	Floating-Point Multiply (Single/Double Precision)
FNEG	Floating-Point Negate
FSNEG, FDNEG	Floating-Point Negate (Single/Double Precision)
FNOP	Floating-Point No Operation

Table A-2. M68000 Family Instruction Set (Continued)

Mnemonic	Description
FREM	IEEE Remainder
FRESTORE	Restore Floating-Point Internal State
FSAVE	Save Floating-Point Internal State
FSCALE	Floating-Point Scale Exponent
FScC	Floating-Point Set According to Condition
FSGLDIV	Single-Precision Divide
FSGLMUL	Single-Precision Multiply
FSIN	Sine
FSINCOS	Simultaneous Sine and Cosine
FSINH	Hyperbolic Sine
FSQRT	Floating-Point Square Root
FSSQRT,FDSQRT	Floating-Point Square Root (Single/Double Precision)
FSUB	Floating-Point Subtract
FSSUB,FDSUB	Floating-Point Subtract (Single/Double Precision)
FTAN	Tangent
FTANH	Hyperbolic Tangent
FTENTOX	Floating-Point 10x
FTRAPcc	Floating-Point Trap On Condition
FTST	Floating-Point Test
FTWOTOX	Floating-Point 2x
ILLEGAL	Take Illegal Instruction Trap
JMP	Jump
JSR	Jump to Subroutine
LEA	Load Effective Address
LINK	Link and Allocate
LPSTOP	Low-Power Stop
LSL, LSR	Logical Shift Left and Right
MOVE	Move
MOVEA	Move Address
MOVE from CCR	Move from Condition Code Register
MOVE from SR	Move from Status Register
MOVE to CCR	Move to Condition Code Register
MOVE to SR	Move to Status Register
MOVE USP	Move User Stack Pointer
MOVE16	16-Byte Block Move
MOVEC	Move Control Register
MOVEM	Move Multiple Registers
MOVEP	Move Peripheral
MOVEQ	Move Quick
MOVES	Move Alternate Address Space
MULS	Signed Multiply
MULU	Unsigned Multiply
NBCD	Negate Decimal with Extend
NEG	Negate
NEGX	Negate with Extend
NOP	No Operation
NOT	Logical Complement

Table A-2. M68000 Family Instruction Set (Concluded)

Mnemonic	Description
OR	Logical Inclusive-OR
ORI	Logical Inclusive-OR Immediate
ORI to CCR	Inclusive-OR Immediate to Condition Code Register
ORI to SR	Inclusive-OR Immediate to Status Register
PACK	Pack BCD
PBcc	Branch on PMMU Condition
PDBcc	Test, Decrement, and Branch on PMMU Condition
PEA	Push Effective Address
PFLUSH	Flush Entry(ies) in the ATCs
PFLUSHA	Flush Entry(ies) in the ATCs
PFLUSHR	Flush Entry(ies) in the ATCs and RPT Entries
PFLUSHS	Flush Entry(ies) in the ATCs
PLOAD	Load an Entry into the ATC
PMOVE	Move PMMU Register
PRESTORE	PMMU Restore Function
PSAVE	PMMU Save Function
PScC	Set on PMMU Condition
PTEST	Test a Logical Address
PTRAPcc	Trap on PMMU Condition
PVALID	Validate a Pointer
RESET	Reset External Devices
ROL, ROR	Rotate Left and Right
ROXL, ROXR	Rotate with Extend Left and Right
RTD	Return and Deallocate
RTE	Return from Exception
RTM	Return from Module
RTR	Return and Restore
RTS	Return from Subroutine
SBCD	Subtract Decimal with Extend
ScC	Set Conditionally
STOP	Stop
SUB	Subtract
SUBA	Subtract Address
SUBI	Subtract Immediate
SUBQ	Subtract Quick
SUBX	Subtract with Extend
SWAP	Swap Register Words
TAS	Test Operand and Set
TBLS, TBLSN	Signed Table Lookup with Interpolate
TBLU, TBLUN	Unsigned Table Lookup with Interpolate
TRAP	Trap
TRAPcc	Trap Conditionally
TRAPV	Trap on Overflow
TST	Test Operand
UNLK	Unlink
UNPK	Unpack BCD

A.1 MC68000, MC68008, MC68010 PROCESSORS

The following paragraphs provide information on the MC68000, MC68008, and MC68010 instruction set and addressing modes.

A.1.1 M68000, MC68008, and MC68010 Instruction Set

Table A-3 lists the instructions used with the MC68000 and MC68008 processors, and Table A-4 lists the instructions used with MC68010.

Table A-3. MC68000 and MC68008 Instruction Set

Mnemonic	Description
ABCD	Add Decimal with Extend
ADD	Add
ADDA	Add Address
ADDI	Add Immediate
ADDQ	Add Quick
ADDX	Add with Extend
AND	Logical AND
ANDI	Logical AND Immediate
ANDI to CCR	AND Immediate to Condition Code Register
ANDI to SR	AND Immediate to Status Register
ASL, ASR	Arithmetic Shift Left and Right
Bcc	Branch Conditionally
BCHG	Test Bit and Change
BCLR	Test Bit and Clear
BRA	Branch
BSET	Test Bit and Set
BSR	Branch to Subroutine
BTST	Test Bit
CHK	Check Register Against Bound
CLR	Clear
CMP	Compare
CMPA	Compare Address
CMPI	Compare Immediate
CMPM	Compare Memory to Memory
DBcc	Test Condition, Decrement, and Branch
DIVS	Signed Divide
DIVU	Unsigned Divide
EOR	Logical Exclusive-OR
EORI	Logical Exclusive-OR Immediate
EORI to CCR	Exclusive-OR Immediate to Condition Code Register
EORI to SR	Exclusive-OR Immediate to Status Register
EXG	Exchange Registers
EXT	Sign Extend
ILLEGAL	Take Illegal Instruction Trap
JMP	Jump
JSR	Jump to Subroutine

**Table A-3. MC68000 and MC68008 Instruction Set
(Continued)**

Mnemonic	Description
LEA	Load Effective Address
LINK	Link and Allocate
LSL, LSR	Logical Shift Left and Right
MOVE	Move
MOVEA	Move Address
MOVE to CCR	Move to Condition Code Register
MOVE from SR	Move from Status Register
MOVE to SR	Move to Status Register
MOVE USP	Move User Stack Pointer
MOVEM	Move Multiple Registers
MOVEP	Move Peripheral
MOVEQ	Move Quick
MULS	Signed Multiply
MULU	Unsigned Multiply
NBCD	Negate Decimal with Extend
NEG	Negate
NEGX	Negate with Extend
NOP	No Operation
NOT	Logical Complement
OR	Logical Inclusive-OR
ORI	Logical Inclusive-OR Immediate
ORI to CCR	Inclusive-OR Immediate to Condition Code Register
ORI to SR	Inclusive-OR Immediate to Status Register
PEA	Push Effective Address
RESET	Reset External Devices
ROL, ROR	Rotate Left and Right
ROXL, ROXR	Rotate with Extend Left and Right
RTE	Return from Exception
RTR	Return and Restore
RTS	Return from Subroutine
SBCD	Subtract Decimal with Extend
Scc	Set Conditionally
STOP	Stop
SUB	Subtract
SUBA	Subtract Address
SUBI	Subtract Immediate
SUBQ	Subtract Quick
SUBX	Subtract with Extend
SWAP	Swap Register Words
TAS	Test Operand and Set
TRAP	Trap
TRAPV	Trap on Overflow
TST	Test Operand
UNLK	Unlink

Table A-4. MC68010 Instruction Set

Mnemonic	Description
ABCD	Add Decimal with Extend
ADD	Add
ADDA	Add Address
ADDI	Add Immediate
ADDQ	Add Quick
ADDX	Add with Extend
AND	Logical AND
ANDI	Logical AND Immediate
ANDI to CCR	AND Immediate to Condition Code Register
ANDI to SR	AND Immediate to Status Register
ASL, ASR	Arithmetic Shift Left and Right
Bcc	Branch Conditionally
BCHG	Test Bit and Change
BCLR	Test Bit and Clear
BKPT	Breakpoint
BRA	Branch
BSET	Test Bit and Set
BSR	Branch to Subroutine
BTST	Test Bit
CHK	Check Register Against Bound
CLR	Clear
CMP	Compare
CMPA	Compare Address
CMPI	Compare Immediate
CMPM	Compare Memory to Memory
DBcc	Test Condition, Decrement and Branch
DIVS	Signed Divide
DIVU	Unsigned Divide
EOR	Logical Exclusive-OR
EORI	Logical Exclusive-OR Immediate
EORI to CCR	Exclusive-OR Immediate to Condition Code Register
EORI to SR	Exclusive-OR Immediate to Status Register
EXG	Exchange Registers
EXT	Sign Extend
ILLEGAL	Take Illegal Instruction Trap
JMP	Jump
JSR	Jump to Subroutine

Table A-4. MC68010 Instruction Set (Continued)

Mnemonic	Description
LEA	Load Effective Address
LINK	Link and Allocate
LSL, LSR	Logical Shift Left and Right
MOVE	Move
MOVEA	Move Address
MOVE from CCR	Move from Condition Code Register
MOVE from SR	Move from Status Register
MOVE to CCR	Move to Condition Code Register
MOVE to SR	Move to Status Register
MOVE USP	Move User Stack Pointer
MOVEC	Move Control Register
MOVEM	Move Multiple Registers
MOVEP	Move Peripheral
MOVEQ	Move Quick
MOVES	Move Address Space
MULS	Signed Multiply
MULU	Unsigned Multiply
NBCD	Negate Decimal with Extend
NEG	Negate
NEGX	Negate with Extend
NOP	No Operation
NOT	Logical Complement
OR	Logical Inclusive-OR
ORI	Logical Inclusive-OR Immediate
ORI to CCR	Inclusive-OR Immediate to Condition Code Register
ORI to SR	Inclusive-OR Immediate to Status Register
PEA	Push Effective Address
RESET	Reset External Devices
ROL, ROR	Rotate Left and Right
ROXL, ROXR	Rotate with Extend Left and Right
RTD	Return and Deallocate
RTE	Return from Exception
RTR	Return and Restore
RTS	Return from Subroutine
SBCD	Subtract Decimal with Extend
Scc	Set Conditionally
STOP	Stop
SUB	Subtract
SUBA	Subtract Address
SUBI	Subtract Immediate
SUBQ	Subtract Quick
SUBX	Subtract with Extend
SWAP	Swap Register Words
TAS	Test Operand and Set
TRAP	Trap
TRAPV	Trap on Overflow
TST	Test Operand
UNLK	Unlink

A.1.2 MC68000, MC68008, and MC68010 Addressing Modes

The MC68000, MC68008, and MC68010 support 14 addressing modes as shown in Table A-5.

**Table A-5. MC68000, MC68008, and MC68010
Data Addressing Modes**

Mode	Generation
Register Direct Addressing Data Register Direct Address Register Direct	<ea> = Dn <ea> = An
Absolute Data Addressing Absolute Short Absolute Long	<ea> = (Next Word) <ea> = (Next Two Words)
Program Counter Relative Addressing Relative with Offset Relative with Index and Offset	<ea> = (PC) + d ₁₆ <ea> = (PC) + d ₈
Register Indirect Addressing Register Indirect Postincrement Register Indirect Predecrement Register Indirect Register Indirect with Offset Indexed Register Indirect with Offset	<ea> = (An) <ea> = (An), An - An + N An - An - N, <ea> = (An) <ea> = (An) + d ₁₆ <ea> = (An) + (Xn) + d ₈
Immediate Data Addressing Immediate Quick Immediate	DATA = Next Word(s) Inherent Data
Implied Addressing Implied Register	<ea> = SR, USP, SSP, PC, VBR, SFC, DFC

N = 1 for byte, 2 for word, and 4 for long word. If An is the stack pointer and the operand size is byte, N = 2 to keep the stack pointer on a word boundary.

A.2 MC68020 PROCESSORS

The following paragraphs provide information on the MC68020 instruction set and addressing modes.

A.2.1 MC68020 Instruction Set

Table A-6 lists the instructions used with the MC68020 processors.

Table A-6. MC68020 Instruction Set Summary

Mnemonic	Description
ABCD	Add Decimal with Extend
ADD	Add
ADDA	Add Address
ADDI	Add Immediate
ADDQ	Add Quick
ADDX	Add with Extend
AND	Logical AND
ANDI	Logical AND Immediate
ANDI to CCR	AND Immediate to Condition Code Register
ANDI to SR	AND Immediate to Status Register
ASL, ASR	Arithmetic Shift Left and Right
Bcc	Branch Conditionally
BCHG	Test Bit and Change
BCLR	Test Bit and Clear
BFCHG	Test Bit Field and Change
BFCLR	Test Bit Field and Clear
BFEXTS	Signed Bit Field Extract
BFEXTU	Unsigned Bit Field Extract
BFFFO	Bit Field Find First One
BFINS	Bit Field Insert
BFSET	Test Bit Field and Set
BFTST	Test Bit Field
BKPT	Breakpoint
BRA	Branch
BSET	Test Bit and Set
BSR	Branch to Subroutine
BTST	Test Bit
CALLM	CALL Module
CAS	Compare and Swap Operands
CAS2	Compare and Swap Dual Operands
CHK	Check Register Against Bound
CHK2	Check Register Against Upper and Lower Bounds
CLR	Clear
CMP	Compare
CMP2	Compare Register Against Upper and Lower Bounds
CMPA	Compare Address
CMPI	Compare Immediate

**Table A-6. MC68020 Instruction Set Summary
(Continued)**

Mnemonic	Description
CMPM	Compare Memory to Memory
cpBcc	Branch to Coprocessor Condition
cpDBcc	Test Coprocessor Condition, Decrement and Branch
cpGEN	Coprocessor General Function
cpRESTORE	Coprocessor Restore Function
cpSAVE	Coprocessor Save Function
cpScc	Set on Coprocessor Condition
cpTRACPcc	Trap on Coprocessor Condition
DBcc	Test Condition, Decrement, and Branch
DIVS, DIVSL	Signed Divide
DIVU, DIVUL	Unsigned Divide
EOR	Logical Exclusive-OR
EORI	Logical Exclusive-OR Immediate
EORI to CCR	Exclusive-OR Immediate to Condition Code Register
EORI to SR	Exclusive-OR Immediate to Status Register
EXG	Exchange Registers
EXT, EXTB	Sign Extend
ILLEGAL	Take Illegal Instruction Trap
JMP	Jump
JSR	Jump to Subroutine
LEA	Load Effective Address
LINK	Link and Allocate
LSL, LSR	Logical Shift Left and Right
MOVE	Move
MOVEA	Move Address
MOVE from CCR	Move from Condition Code Register
MOVE from SR	Move from Status Register
MOVE to CCR	Move to Condition Code Register
MOVE to SR	Move to Status Register
MOVE USP	Move User Stack Pointer
MOVEC	Move Control Register
MOVEM	Move Multiple Registers
MOVEP	Move Peripheral
MOVEQ	Move Quick
MOVES	Move Alternate Address Space
MULS	Signed Multiply
MULU	Unsigned Multiply
NBCD	Negate Decimal with Extend
NEG	Negate
NEGX	Negate with Extend
NOP	No Operation
NOT	Logical Complement

**Table A-6. MC68020 Instruction Set Summary
(Concluded)**

Mnemonic	Description
OR	Logical Inclusive-OR
ORI	Logical Inclusive-OR Immediate
ORI to CCR	Inclusive-OR Immediate to Condition Code Register
ORI to SR	Inclusive-OR Immediate to Status Register
PACK	Pack BCD
PEA	Push Effective Address
RESET	Reset External Devices
ROL, ROR	Rotate Left and Right
ROXL, ROXR	Rotate with Extend Left and Right
RTD	Return and Deallocate
RTE	Return from Exception
RTM	Return from Module
RTR	Return and Restore
RTS	Return from Subroutine
SBCD	Subtract Decimal with Extend
Scc	Set Conditionally
STOP	Stop
SUB	Subtract
SUBA	Subtract Address
SUBI	Subtract Immediate
SUBQ	Subtract Quick
SUBX	Subtract with Extend
SWAP	Swap Register Words
TAS	Test Operand and Set
TRAP	Trap
TRAPcc	Trap Conditionally
TRAPV	Trap on Overflow
TST	Test Operand
UNLK	Unlink
UNPK	Unpack BCD

A.2.2 MC68020 Addressing Modes

The MC68020 supports 18 addressing modes as shown in Table A-7.

Table A-7. MC68020 Data Addressing Modes

Addressing Modes	Syntax
Register Direct Address Register Direct Address Register Direct	Dn An
Register Indirect Address Register Indirect Address Register Indirect with Postincrement Address Register Indirect with Predecrement Address Register Indirect with Displacement	(An) (An)+ -(An) (d ₁₆ ,An)
Register Indirect with Index Address Register Indirect with Index (8-Bit Displacement) Address Register Indirect with Index (Base Displacement)	(d ₈ ,An,Xn) (bd,An,Xn)
Memory Indirect Memory Indirect Postindexed Memory Indirect Preindexed	([bd,An],Xn,od) ([bd,An,Xn],od)
Program Counter Indirect with Displacement	(d ₁₆ ,PC)
Program Counter Indirect with Index PC Indirect with Index (8-Bit Displacement) PC Indirect with Index (Base Displacement)	(d ₈ ,PC,Xn) (bd,PC,Xn)
Program Counter Memory Indirect PC Memory Indirect Postindexed PC Memory Indirect Preindexed	([bd,PC],Xn,od) ([bd,PC,Xn],od)
Absolute Absolute Short Absolute Long	(xxx).W (xxx).L
Immediate	#<data>

A.3 MC68030 PROCESSORS

The following paragraphs provide information on the MC68030 instruction set and addressing modes.

A.3.1 MC68030 Instruction Set

Table A-8 lists the instructions used with the MC68030 processors.

Table A-8. MC68030 Instruction Set Summary

Mnemonic	Description
ABCD	Add Decimal with Extend
ADD	Add
ADDA	Add Address
ADDI	Add Immediate
ADDQ	Add Quick
ADDX	Add with Extend
AND	Logical AND
ANDI	Logical AND Immediate
ANDI to CCR	AND Immediate to Condition Code Register
ANDI to SR	AND Immediate to Status Register
ASL, ASR	Arithmetic Shift Left and Right
Bcc	Branch Conditionally
BCHG	Test Bit and Change
BCLR	Test Bit and Clear
BFCHG	Test Bit Field and Change
BFCLR	Test Bit Field and Clear
BFEXTS	Signed Bit Field Extract
BFEXTU	Unsigned Bit Field Extract
BFFFO	Bit Field Find First One
BFINS	Bit Field Insert
BFSET	Test Bit Field and Set
BFTST	Test Bit Field
BKPT	Breakpoint
BRA	Branch
BSET	Test Bit and Set
BSR	Branch to Subroutine
BTST	Test Bit
CAS	Compare and Swap Operands
CAS2	Compare and Swap Dual Operands
CHK	Check Register Against Bound
CHK2	Check Register Against Upper and Lower Bounds
CLR	Clear
CMP	Compare
CMPA	Compare Address
CMPI	Compare Immediate
CMPM	Compare Memory to Memory

**Table A-8. MC68030 Instruction Set Summary
(Continued)**

Mnemonic	Description
CMP2	Compare Register Against Upper and Lower Bounds
cpBcc	Branch on Coprocessor Condition
cpDBcc	Test Coprocessor Condition, Decrement and Branch
cpGEN	Coprocessor General Function
cpRESTORE	Coprocessor Restore Function
cpSAVE	Coprocessor Save Function
cpScc	Set on Coprocessor Condition
cpTRAPcc	Trap on Coprocessor Condition
DBcc	Test Condition, Decrement and Branch
DIVS, DIVSL	Signed Divide
DIVU, DIVUL	Unsigned Divide
EOR	Logical Exclusive-OR
EORI	Logical Exclusive-OR Immediate
EORI to CCR	Exclusive-OR Immediate to Condition Code Register
EORI to SR	Exclusive-OR Immediate to Status Register
EXG	Exchange Registers
EXT, EXTB	Sign Extend
ILLEGAL	Take Illegal Instruction Trap
JMP	Jump
JSR	Jump to Subroutine
LEA	Load Effective Address
LINK	Link and Allocate
LSL, LSR	Logical Shift Left and Right
MOVE	Move
MOVEA	Move Address
MOVE from CCR	Move from Condition Code Register
MOVE to CCR	Move to Condition Code Register
MOVE from SR	Move from Status Register
MOVE to SR	Move to Status Register
MOVE USP	Move User Stack Pointer
MOVEC	Move Control Register
MOVEM	Move Multiple Registers
MOVEP	Move Peripheral
MOVEQ	Move Quick
MOVES	Move Alternate Address Space
MULS	Signed Multiply
MULU	Unsigned Multiply
NBCD	Negate Decimal with Extend
NEG	Negate
NEGX	Negate with Extend
NOP	No Operation
NOT	Logical Complement

**Table A-8. MC68030 Instruction Set Summary
(Concluded)**

Mnemonic	Description
NBCD	Negate Decimal with Extend
NEG	Negate
NEGX	Negate with Extend
NOP	No Operation
NOT	Logical Complement
OR	Logical Inclusive-OR
ORI	Logical Inclusive-OR Immediate
ORI to CCR	Inclusive-OR Immediate to Condition Code Register
ORI to SR	Inclusive-OR Immediate to Status Register
PACK	Pack BCD
PEA	Push Effective Address
PFLUSH*	Invalidate Entries in the ATC
PFLUSHA*	Invalidate all Entries in the ATC
PLOAD*	Load an Entry into the ATC
PMOVE	Move PMMU Register
PTEST	Get Information about Logical Address
RESET	Reset External Devices
ROL, ROR	Rotate Left and Right
ROXL, ROXR	Rotate with Extend Left and Right
RTD	Return and Deallocate
RTE	Return from Exception
RTR	Return and Restore
RTS	Return from Subroutine
SBCD	Subtract Decimal with Extend
Scc	Set Conditionally
STOP	Stop
SUB	Subtract
SUBA	Subtract Address
SUBI	Subtract Immediate
SUBQ	Subtract Quick
SUBX	Subtract with Extend
SWAP	Swap Register Words
TAS	Test Operand and Set
TRAP	Trap
TRAPcc	Trap Conditionally
TRAPV	Trap on Overflow
TST	Test Operand
UNLK	Unlink
UNPK	Unpack BCD

*Not applicable to the MC68EC030

A.3.2 MC68030 Addressing Modes

The MC68030 supports 18 addressing modes as shown in Table A-9.

Table A-9. MC68030 Data Addressing Modes

Addressing Modes	Syntax
Register Direct Data Register Direct Address Register Direct	Dn An
Register Indirect Address Register Indirect Address Register Indirect with Postincrement Address Register Indirect with Predecrement Address Register Indirect with Displacement	(An) (An)+ -(An) (d ₁₆ ,An)
Register Indirect with Index Address Register Indirect with Index (8-Bit Displacement) Address Register Indirect with Index (Base Displacement)	(d ₈ ,An,Xn) (bd,An,Xn)
Memory Indirect Memory Indirect Postindexed Memory Indirect Preindexed	([bd,An],Xn,od) ([bd,An,Xn],od)
Program Counter Indirect with Displacement	(d ₁₆ ,PC)
Program Counter Indirect with Index PC Indirect with Index (8-Bit Displacement) PC Indirect with Index (Base Displacement)	(d ₈ ,PC,Xn) (bd,PC,Xn)
Program Counter Memory Indirect PC Memory Indirect Postindexed PC Memory Indirect Preindexed	([bd,PC],Xn,od) ([bd,PC,Xn],od)
Absolute Absolute Short Absolute Long	(xxx).W (xxx).L
Immediate	#<data>

A.4 MC68040 PROCESSORS

The following paragraphs provide information on the MC68040 instruction set and addressing modes.

A.4.1 MC68040 Instruction Set

Table A-10 lists the instructions used with the MC68040 processor.

Table A-10. MC68040 Instruction Set

Mnemonic	Description
ABCD	Add Decimal with Extend
ADD	Add
ADDA	Add Address
ADDI	Add Immediate
ADDQ	Add Quick
ADDX	Add with Extend
AND	Logical AND
ANDI	Logical AND Immediate
ANDI to CCR	AND Immediate to Condition Code Register
ANDI to SR	AND Immediate to Status Register
ASL, ASR	Arithmetic Shift Left and Right
Bcc	Branch Conditionally
BCHG	Test Bit and Change
BCLR	Test Bit and Clear
BFCHG	Test Bit Field and Change
BFCLR	Test Bit Field and Clear
BFEXTS	Signed Bit Field Extract
BFEXTU	Unsigned Bit Field Extract
BFFFO	Bit Field Find First One
BFINS	Bit Field Insert
BFSET	Test Bit Field and Set
BFTST	Test Bit Field
BKPT	Breakpoint
BRA	Branch
BSET	Test Bit and Set
BSR	Branch to Subroutine
BTST	Test Bit
CAS	Compare and Swap Operands
CAS2	Compare and Swap Dual Operands
CHK	Check Register Against Bound
CHK2	Check Register Against Upper and Lower Bounds
CINV	Invalidate Cache Entries
CLR	Clear
CMP	Compare
CMPA	Compare Address

Table A-10. MC68040 Instruction Set (Continued)

Mnemonic	Description
CMPI	Compare Immediate
CMPM	Compare Memory to Memory
CMP2	Compare Register Against Upper and Lower Bounds
CPUSH	Push then Invalidate Cache Entries
DBcc	Test Condition, Decrement and Branch
DIVS, DIVSL	Signed Divide
DIVU, DIVUL	Unsigned Divide
EOR	Logical Exclusive-OR
EORI	Logical Exclusive-OR Immediate
EORI to CCR	Exclusive-OR Immediate to Condition Code Register
EORI to SR	Exclusive-OR Immediate to Status Register
EXG	Exchange Registers
EXT, EXTB	Sign Extend
FABS ¹	Floating-Point Absolute Value
FSABS, FDABS ¹	Floating-Point Absolute Value (Single/Double Precision)
FACOS ^{1,2}	Floating-Point Arc Cosine
FADD ¹	Floating-Point Add
FSADD, FDADD ¹	Floating-Point Add (Single/Double Precision)
FASIN ^{1,2}	Floating-Point Arc Sine
FATAN ^{1,2}	Floating-Point Arc Tangent
FATANH ^{1,2}	Floating-Point Hyperbolic Arc Tangent
FBcc ¹	Floating-Point Branch
FCMP ¹	Floating-Point Compare
FCOS ^{1,2}	Floating-Point Cosine
FCOSH ^{1,2}	Floating-Point Hyperbolic Cosine
FDBcc ¹	Floating-Point Decrement and Branch
FDIV ¹	Floating-Point Divide
FSDIV, FDDIV ¹	Floating-Point Divide (Single/Double Precision)
FETOX ^{1,2}	Floating-Point e^x
FETOXM1 ^{1,2}	Floating-Point $e^x - 1$
FGETEXP ^{1,2}	Floating-Point Get Exponent
FGETMAN ^{1,2}	Floating-Point Get Mantissa
FINT ^{1,2}	Floating-Point Integer Part
FINTRZ ^{1,2}	Floating-Point Integer Part, Round-to-Zero
FLOG10 ^{1,2}	Floating-Point $\log_{10}$
FLOG2 ^{1,2}	Floating-Point $\log_2$
FLOGN ^{1,2}	Floating-Point $\log_e$
FLOGNP1 ^{1,2}	Floating-Point $\log_e (x + 1)$
FMOD ^{1,2}	Floating-Point Modulo Remainder
FMOVE ¹	Move Floating-Point Register
FSMOVE, FDMOVE ¹	Move Floating-Point Register (Single/Double Precision)
FMOVECR ¹	Move Constant ROM
FMOVEM ¹	Move Multiple Floating-Point Registers
FMUL ¹	Floating-Point Multiply
FSMUL, FDMUL ¹	Floating-Point Multiply (Single/Double Precision)

Table A-10. MC68040 Instruction Set (Continued)

Mnemonic	Description
FNEG ¹	Floating-Point Negate
FSNEG, FDNEG ¹	Floating-Point Negate (Single/Double Precision)
FNOP ¹	Floating-Point No Operation
FREM ^{1,2}	IEEE Remainder
FRESTORE ¹	Restore Floating-Point Internal State
FSAVE ¹	Save Floating-Point Internal State
FSCALE ^{1,2}	Floating-Point Scale Exponent
FSc ¹	Floating-Point Set According to Condition
FSGLDIV ^{1,2}	Single-Precision Divide
FSGLMUL ^{1,2}	Single-Precision Multiply
FSIN ^{1,2}	Sine
FSINCOS ^{1,2}	Simultaneous Sine and Cosine
FSINH ^{1,2}	Hyperbolic Sine
FSQRT ¹	Floating-Point Square Root
FSSQRT, FDSQRT ¹	Floating-Point Square Root (Single/Double Precision)
FSUB ¹	Floating-Point Subtract
FSSUB, FDSUB ¹	Floating-Point Subtract (Single/Double Precision)
FTAN ^{1,2}	Tangent
FTANH ^{1,2}	Hyperbolic Tangent
FTENTOX ^{1,2}	Floating-Point 10 ^x
FTRAPcc ^{1,2}	Floating-Point Trap On Condition
FTST ¹	Floating-Point Test
FTWOTOX ^{1,2}	Floating-Point 2 ^x
ILLEGAL	Take Illegal Instruction Trap
JMP	Jump
JSR	Jump to Subroutine
LEA	Load Effective Address
LINK	Link and Allocate
LSL, LSR	Logical Shift Left and Right
MOVE	Move
MOVEA	Move Address
MOVE from CCR	Move from Condition Code Register
MOVE to CCR	Move to Condition Code Register
MOVE from SR	Move from Status Register
MOVE to SR	Move to Status Register
MOVE USP	Move User Stack Pointer
MOVEC	Move Control Register
MOVEM	Move Multiple Registers
MOVEP	Move Peripheral
MOVEQ	Move Quick
MOVES	Move Alternate Address Space
MOVE16	16-Byte Block Move
MULS	Signed Multiply
MULU	Unsigned Multiply

Table A-10. MC68040 Instruction Set (Concluded)

Mnemonic	Description
NBCD	Negate Decimal with Extend
NEG	Negate
NEGX	Negate with Extend
NOP	No Operation
NOT	Logical Complement
OR	Logical Inclusive-OR
ORI	Logical Inclusive-OR Immediate
ORI to CCR	Inclusive-OR Immediate to Condition Code Register
ORI to SR	Inclusive-OR Immediate to Status Register
PACK	Pack BCD
PEA	Push Effective Address
PFLUSH	Flush Entry(ies) in the ATCs
PFLUSHA	Flush all Entry(ies) in the ATCs
PTEST	Test a Logical Address
RESET	Reset External Devices
ROL, ROR	Rotate Left and Right
ROXL, ROXR	Rotate with Extend Left and Right
RTD	Return and Deallocate
RTE	Return from Exception
RTR	Return and Restore
RTS	Return from Subroutine
SBCD	Subtract Decimal with Extend
Scc	Set Conditionally
STOP	Stop
SUB	Subtract
SUBA	Subtract Address
SUBI	Subtract Immediate
SUBQ	Subtract Quick
SUBX	Subtract with Extend
SWAP	Swap Register Words
TAS	Test Operand and Set
TRAP	Trap
TRAPcc	Trap Conditionally
TRAPV	Trap on Overflow
TST	Test Operand
UNLK	Unlink
UNPK	Unpack BCD

NOTES:

1. Not applicable to the MC68EC040 and MC68LC040.
2. These instructions are software supported.

A.4.2 MC68040 Addressing Modes

The MC68040 supports 18 addressing modes as shown in Table A-11.

Table A-11. MC68040 Data Addressing Modes

Addressing Modes	Syntax
Register Direct Data Register Direct Address Register Direct	Dn An
Register Indirect Address Register Indirect Address Register Indirect with Postincrement Address Register Indirect with Predecrement Address Register Indirect with Displacement	(An) (An) + -(An) (d ₁₆ ,An)
Register Indirect with Index Address Register Indirect with Index (8-Bit Displacement) Address Register Indirect with Index (Base Displacement)	(d ₈ ,An,Xn) (bd,An,Xn)
Memory Indirect Memory Indirect Postindexed Memory Indirect Preindexed	([bd,An],Xn,od) ([bd,An,Xn],od)
Program Counter Indirect with Displacement	(d ₁₆ ,PC)
Program Counter Indirect with Index PC Indirect with Index (8-Bit Displacement) PC Indirect with Index (Base Displacement)	(d ₈ ,PC,Xn) (bd,PC,Xn)
Program Counter Memory Indirect PC Memory Indirect Postindexed PC Memory Indirect Preindexed	([bd,PC],Xn,od) ([bd,PC,Xn],od)
Absolute Absolute Short Absolute Long	xxx.W xxx.L
Immediate	# < data >

A.5 MC68881/MC68882 COPROCESSORS

The following paragraphs provide information on the MC68881/MC68882 instruction set and addressing modes.

A.5.1 MC68881/MC68882 Instruction Set

Table A-12 lists the instructions used with the MC68881/MC68882 coprocessors.

Table A-12. MC68881/MC68882 Instruction Set

Mnemonic	Description
FABS	Floating-Point Absolute Value
FACOS	Floating-Point Arc Cosine
FADD	Floating-Point Add
FASIN	Floating-Point Arc Sine
FATAN	Floating-Point Arc Tangent
FATANH	Floating-Point Hyperbolic Arc Tangent
FBcc	Floating-Point Branch
FCMP	Floating-Point Compare
FCOS	Floating-Point Cosine
FCOSH	Floating-Point Hyperbolic Cosine
FDBcc	Floating-Point Decrement and Branch
FDIV	Floating-Point Divide
FETOX	Floating-Point e^x
FETOXM1	Floating-Point $e^x - 1$
FGETEXP	Floating-Point Get Exponent
FGETMAN	Floating-Point Get Mantissa
FINT	Floating-Point Integer Part
FINTRZ	Floating-Point Integer Part, Round-to-Zero
FLOG10	Floating-Point Log ₁₀
FLOG2	Floating-Point Log ₂
FLOGN	Floating-Point Log _e
FLOGNP1	Floating-Point Log _e ($x + 1$)
FMOD	Floating-Point Modulo Remainder
FMOVE	Move Floating-Point Register
FMOVECR	Move Constant ROM
FMOVEM	Move Multiple Floating-Point Registers
FMUL	Floating-Point Multiply
FNEG	Floating-Point Negate
FNOP	Floating-Point No Operation
FREM	IEEE Remainder
FRESTORE	Restore Floating-Point Internal State
FSAVE	Save Floating-Point Internal State
FSCALE	Floating-Point Scale Exponent
FScc	Floating-Point Set According to Condition
FSGLDIV	Single-Precision Divide
FSGLMUL	Single-Precision Multiply
FSIN	Sine
FSINCOS	Simultaneous Sine and Cosine
FSINH	Hyperbolic Sine

Table A-12. MC68881/MC68882 Instruction Set

Mnemonic	Description
FSQRT	Floating-Point Square Root
FSUB	Floating-Point Subtract
FTAN	Tangent
FTANH	Hyperbolic Tangent
FTENTOX	Floating-Point 10x
FTRAPcc	Floating-Point Trap On Condition
FTST	Floating-Point Test
FTWOTOX	Floating-Point 2x

A.5.2 MC68881/MC68882 Addressing Modes

The MC68881/MC68882 does not perform address calculations. When the floating-point coprocessor instructs the processor to transfer an operand via the coprocessor interface, the processor performs the addressing mode calculation requested in the instruction.

A.6 MC68851 COPROCESSORS

The following paragraphs provide information on the MC68851 instruction set and addressing modes.

A.6.1 MC68851 Instruction Set

Table A-13 lists the instructions used with the MC68851 coprocessor.

Table A-13. MC68851 Instruction Set

Mnemonic	Description
PBcc	Branch on PMMU Condition
PDBcc	Test, Decrement, and Branch on PMMU Condition
PFLUSH	Flush Entry(ies) in the ATCs
PFLUSHA	Flush Entry(ies) in the ATCs
PFLUSHR	Flush Entry(ies) in the ATCs and RPT Entries
PFLUSHS	Flush Entry(ies) in the ATCs
PLOAD	Load an Entry into the ATC
PMOVE	Move PMMU Register
PRESTORE	PMMU Restore Function
PSAVE	PMMU Save Function
PScC	Set on PMMU Condition
PTEST	Test a Logical Address
PTRAPcc	Trap on PMMU Condition
PVALID	Validate a Pointer

A.6.2 MC68851 Addressing Modes

The MC68851 supports the same addressing modes as the MC68020 (see Table A-7).